

Lecture 1 Quiz Questions and Answers

1 What is a system?

combination of components acting together to perform a specific objective

☐ 17%

they are sets of connected objects or things

☐ 0%

behaviors of these systems are shaped by their environments

☐ 0%

1 and 3

☐ 22%

All of the above ☒

☐

2 Select the correct statement

dynamic systems can only be found in mechanical field

☐ 3%

we can use physical laws to understand the performance of dynamic systems ☒

☐

the output of a system is only dependent on the input

☐ 0%

All of the above

☐ 26%

None of the above

☐ 6%

3

Select the correct statement

force applied to a damper is inversely proportional to the velocity of the piston

☐ 0%

dynamic systems can be modelled in mathematical terms

☐ 9%

model can be obtained by testing a prototype of a corresponding device

☐ 0%

1 and 2

☐ 9%

2 and 3 ☒

☒

4

Select the correct statement

depending on the time constant, system reach to its steady state at different times ☒

☒ 36%

natural frequency, which describes the speed of the input

☐ 4%

when system is overdamped, it shows both oscillatory and exponential behaviour

☐ 14%

2 and 3

☐

None of the above

☐ 4%

5

Select the correct statement

peak time is the time required for the response to reach steady state value for the first time

 13%

settling time is the time required for the response to reach its steady state value



rise time is the time taken to maximum overshoot

 13%

All of the above

 34%

None of the above ☒

 3%